Package ‘rENM.core’

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Type Package

Title Shared utilities for the rENM framework

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Description The rENM framework is a modular sysem for reconstucting and analyzing long-term ecological niche dynamics using time-indexed environmental data and species occurrence records. Its packages support climate-based niche modeling, temporal analysis of climatic suitability, and reproducible workflows for studying ecological change.

Depends R (>= 4.1.0)

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URL https://github.com/rENM-Framework/rENM.core

BugReports https://github.com/rENM-Framework/rENM.core/issues

Contents

get_species_info — *Look up species metadata by four-letter alpha code*

Description

Retrieves a single species record from the rENM species metadata table stored within the project directory using a standardized alpha code lookup.

Usage

```
get_species_info(alpha_code, project_dir = NULL)
```

Arguments

`alpha_code` Character. Four-letter species alpha code (e.g., "CASP"). Matching is case-insensitive.

`project_dir` Character. Optional rENM project root directory. If NULL, the directory is resolved using `rENM_project_dir` using the following precedence:

1. Explicit project_dir argument
2. options(rENM.project_dir)
3. RENM_PROJECT_DIR environment variable

Supplying project_dir explicitly is recommended for reproducible scripts and for package tests.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context The project directory is resolved using `rENM_project_dir`, which centralizes configuration for all filesystem access in the rENM package and avoids hard-coded paths such as ~/rENM. This design improves portability, reproducibility, and CRAN compliance.

Inputs The species table is expected at:

<rENM_project_dir>/data/_species.csv

The lookup uses a four-letter alpha code (e.g., eBird or USGS banding codes) and is case-insensitive. The function tolerates variation in column names (e.g., COMMON.NAME vs Common Name, SCIENTIFIC.NAME vs SciName, ALPHA.CODE vs SPECIESCODE).

The species table must contain a column representing the four-letter alpha code. The function attempts to match any of the following (case-insensitive, punctuation ignored):

- ALPHA.CODE
- SPECIESCODE
- BANDINGCODE
- ALPHACODE
- CODE

Outputs Returned column names are standardized regardless of the source file.

The returned data frame always contains these columns:

- COMMON.NAME
- SCIENTIFIC.NAME
- EBD.RECORDS
- EBD.RANGE
- GAP.RANGE

Methods Column-name normalization is performed by converting names to uppercase and removing punctuation so that variations such as COMMON.NAME, Common Name, and common_name resolve to the same comparison key.

Data requirements The species table must exist, be readable, and contain at least one row with the required columns and a valid alpha code field.

Value

Data frame. Returns a one-row data frame containing standardized species metadata:

COMMON.NAME Character. Common (English) species name.

SCIENTIFIC.NAME Character. Binomial scientific name.

EBD.RECORDS Numeric. Approximate number of eBird occurrence records.

EBD.RANGE Character. Range description from eBird metadata.

GAP.RANGE Character. Distribution metadata from USGS GAP analysis.

Side effects:

- Reads the species metadata table from disk
- Validates column structure and content
- Standardizes output column names

Examples

```
## Not run:

# Standard lookup
info <- get_species_info("CASP")

# Reproducible script with explicit project directory
info <- get_species_info("CASP", project_dir = "/projects/rENM")


## End(Not run)
```

`rENM_project_dir` *Resolve and validate an rENM project directory on the local filesystem*

Description

Determines the rENM project root directory using a defined precedence order and verifies that it exists, returning a normalized absolute path suitable for use by package functions.

Usage

```
rENM_project_dir(project_dir = NULL)
```

Arguments

`project_dir`	Character. Optional path to the rENM project root. If provided, it takes precedence over options and environment variables.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context This helper determines the rENM project root directory and verifies that it exists. It is intended to centralize project-directory configuration for all functions in the package, ensuring CRAN-friendly behavior with no hard-coded home paths, no assumptions about working directory, and clear error messages.

The project directory is resolved using the following precedence:

1. The explicit project_dir argument (recommended for scripts/tests)
2. options(rENM.project_dir = "...")
3. The environment variable RENM_PROJECT_DIR

If a candidate value is found, it is expanded (e.g., "~"), normalized to an absolute path, and verified to exist using normalizePath(..., mustWork = TRUE). If no valid directory can be determined, the function errors with actionable setup instructions.

Inputs The function accepts an optional project directory path or attempts to resolve it from configuration sources including options and environment variables.

Outputs Returns a normalized absolute path to an existing directory.

Methods Resolution is performed using a helper that:

- Validates that input is a non-empty scalar character string
- Expands "~" using path.expand
- Normalizes and verifies existence using normalizePath with mustWork = TRUE

If a candidate fails validation or does not exist, it is ignored and the next configuration source is attempted.

Data requirements The resolved directory must exist on the local filesystem. This function does not validate directory contents (e.g., presence of data/_species.csv); content validation is the responsibility of calling functions.

Configuration Users can configure the project directory in several ways:

- Session-only: `Sys.setenv(RENM_PROJECT_DIR = "~/rENM")`
- Session-only: `options(rENM.project_dir = "~/rENM")`
- Persistent: add `RENM_PROJECT_DIR=~/rENM` to `~/.Renviron`

In package examples and tests, it is best practice to pass project_dir explicitly (e.g., a temporary directory or `system.file("extdata", package = "rENM")`).

Value

Character. A length-1 string representing the normalized absolute path to the existing project directory.

Side effects:

- Expands and normalizes filesystem paths
- Validates existence of the resolved directory
- Throws an error if no valid directory can be resolved

See Also

normalizePath

Examples

```
## Not run:
# Preferred for reproducible scripts:
rENM_project_dir("/projects/rENM")

# Convenience for interactive use:
Sys.setenv(RENM_PROJECT_DIR = "~/rENM")
rENM_project_dir()

# Or using an R option:
options(rENM.project_dir = "~/rENM")
rENM_project_dir()

## End(Not run)
```

---

show_species *Show available species definitions in the data directory*

Description

Reads and displays the contents of the species metadata table to support inspection and validation of species definitions used in the rENM workflow.

Usage

```
show_species()
```

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context This function is intended as a simple inspection utility to verify species metadata (e.g., alpha codes, GAP ranges, names) used throughout the rENM workflow.

Inputs The species file is expected at:

<project_dir>/data/_species.csv

The CSV file is expected to contain metadata describing species used in the modeling workflow (e.g., ALPHA.CODE, GAP.RANGE, species name). No strict schema is enforced, but column headers are assumed to be present.

Outputs The function prints a formatted summary and the full contents of the species table to the console with left-justified columns.

Methods This function performs:

- Resolution of the project root directory using `rENM_project_dir`
- Location of the species file in the data directory
- Reading the CSV file using `utils::read.csv`
- Printing a formatted header and table contents to the console

Data requirements The file must exist and be readable as a CSV file. If the file does not exist or cannot be read, the function stops with a clear error message.

Console output The function prints:

- A header block with timestamp and file path
- The number of rows and columns detected
- The full data frame with left-justified columns

This function does not write to the run log (`_log.txt`) because it is intended as a lightweight inspection tool.

Value

Data frame. Invisibly returns the data frame read from `_species.csv`.

Side effects:

- Prints formatted output to the console
- Validates file existence and readability

See Also

rENM_project_dir

Examples

```
## Not run:
  # Display all species defined for the project
  sp <- show_species()

## End(Not run)
```

`show_variables` *Show available variable definitions in the data directory*

Description

Reads and displays the contents of the variables metadata table to support inspection and validation of environmental variables used in the rENM workflow.

Usage

```
show_variables()
```

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context This function is intended as a simple inspection utility to verify available environmental variables (e.g., MERRA-2 or derived variables) used throughout the rENM workflow.

Inputs The variables file is expected at:

`<project_dir>/data/_variables.csv`

The CSV file is expected to contain metadata describing environmental variables (e.g., variable names, descriptions, units, sources). No strict schema is enforced, but column headers are assumed to be present.

Outputs The function prints a formatted summary and the full contents of the variables table to the console with left-justified columns.

Methods This function performs:

- Resolution of the project root directory using `rENM_project_dir`
- Location of the variables file in the data directory
- Reading the CSV file using `utils::read.csv`
- Printing a formatted header and table contents to the console

Data requirements The file must exist and be readable as a CSV file. If the file does not exist or cannot be read, the function stops with a clear error message.

Console output The function prints:

- A header block with timestamp and file path
- The number of rows and columns detected
- The full data frame with left-justified columns

This function does not write to the run log (`_log.txt`) because it is intended as a lightweight inspection tool.

Value

Data frame. Invisibly returns the data frame read from `_variables.csv`.

Side effects:

- Prints formatted output to the console
- Validates file existence and readability

See Also

rENM_project_dir

Examples

```
## Not run:
  # Display all variables defined for the project
  vars <- show_variables()

## End(Not run)
```

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